

7 (a) (i) $Q = (M_{Ra} - M_{Rn} - M_{\alpha})c^2$
 $= (226.0254 - 222.0176 - 4.0026) \times (3.00 \times 10^8)^2$
 $= 7.7688 \times 10^{-13} \text{ J}$ **C1**
 $= (7.7688 \times 10^{-13}) / (10^6 \times 1.6 \times 10^{-19})$ **M1**
 $= 4.86 \text{ MeV}$ **A0**

(ii) Conservation of momentum: $MV + M_{\alpha} V_{\alpha} = 0$ (1) **M1**

Conservation of energy: $Q = \frac{1}{2}MV^2 + \frac{1}{2}M_{\alpha}V_{\alpha}^2$ (2) **M1**

From (1): $V = -\left(\frac{M_{\alpha}}{M}\right)V_{\alpha}$ (3)

Sub (3) into (2)

$$Q = \frac{1}{2}M\left(-\left(\frac{M_{\alpha}}{M}\right)V_{\alpha}\right)^2 + \frac{1}{2}M_{\alpha}V_{\alpha}^2$$

$$= \frac{1}{2}M_{\alpha}V_{\alpha}^2\left(\frac{M_{\alpha}}{M} + 1\right)$$
 M1

$$= K_{\alpha}\left(\frac{M_{\alpha}}{M} + 1\right)$$
 A0

(iii) 1. $4.86 = K_{\alpha}\left(\frac{4.0026}{222.0176} + 1\right)$ **C1**

$K_{\alpha} = 4.77 \text{ MeV}$ **A1**

2. The alpha particles carries away most of the energy (98%). **B1**

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(b) alpha particle kinetic energy per mm = $\frac{4.77}{25}$
= 0.191 MeV

C1

energy to produce an ion pair = $\frac{0.191 \times 1.6 \times 10^{-13}}{5.0 \times 10^3}$

C1

= 6.1×10^{-18} J

A1

Section B

8 (a) (i) $2mv$

A1